Section Report 2 — WGS: Reads to Info

BIOL 4810

2026-03-12

What you are doing: You are given **raw reads only**. You must run the full WGS workflow and document it in a way that is reproducible and ready to commit to Git.
Submission: Fill this page out, upload the requested screenshots/PDFs into the page, then use **Download as PDF (Print)**.
Notes: You should taking detailed notes about each step, what software you are using, what version of the software (if available), and output files so that someone else could theoretically repeat your exact work.

---

0) Required folder structure (make this first)

Inside your course repository, create a folder named something like:

```
sectionreport2_wgs/
```

Inside that folder, create:

```
rawfastqc/
fastp/
trimmedfastqc/
assembly/
quast/
filteredassembly/
checkm2/
prokka/
ani/
tygs/
miningfiles/
```

---

1) FASTQC on raw reads (Galaxy)

In Galaxy 1. Upload your raw reads (make sure to create the paired data set) 2. Run **FastQC** on **raw** reads 3. Open the FastQC HTML report

Download requirement: Download the FastQC HTML report for **either R1 or R2** and save it in: `rawfastqc/`

1A. FastQC version (from Galaxy Job Info):

Galaxy Version 0.74+galaxy1

1B. Read length reported:

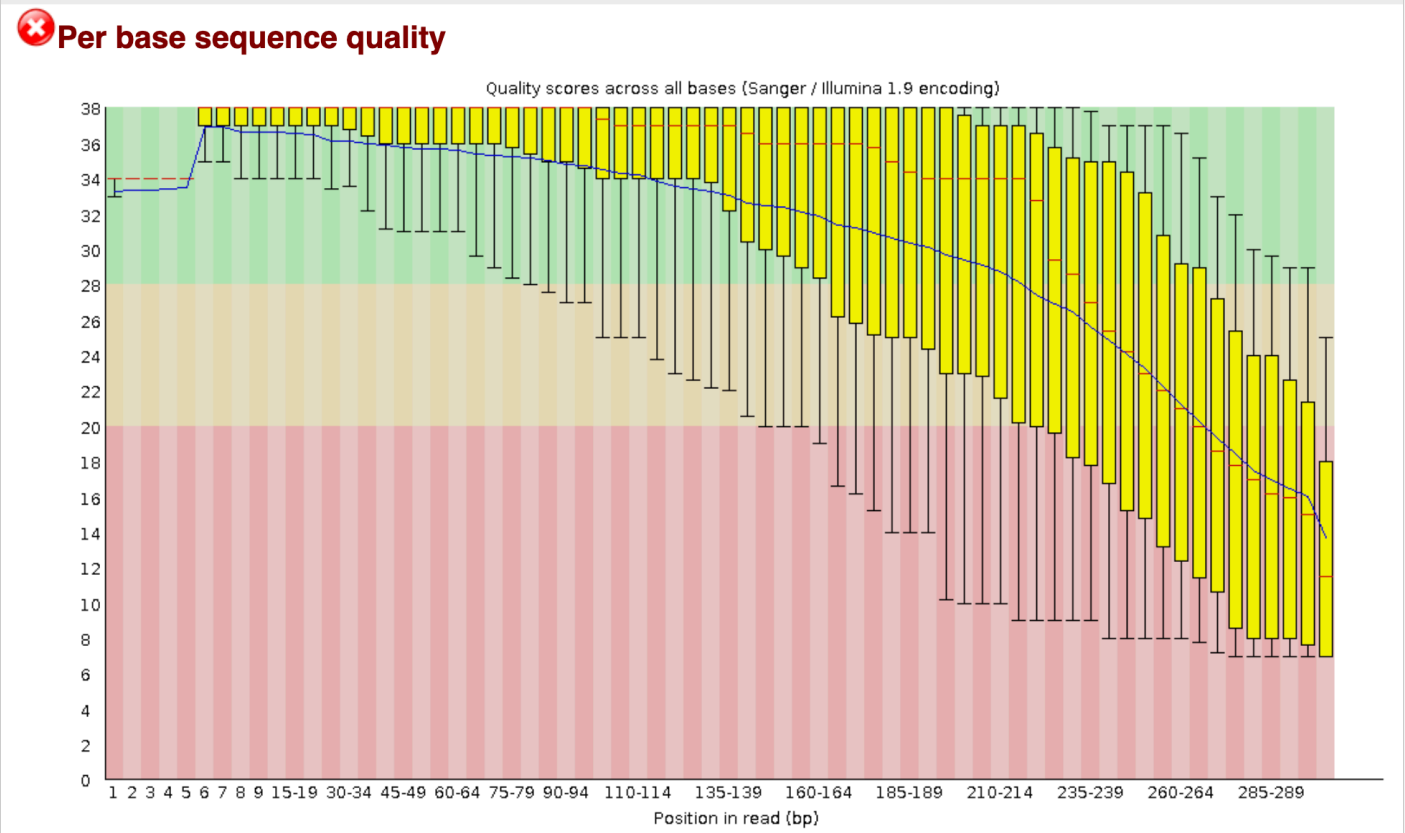
301 bp

1C. One key quality issue you see in raw reads (2–3 sentences):

One key issue seen in the raw reads is a drop in the per base quality, especially when moving towards the end of the reads. This decline in the per base sequence quality plot can be problematic because in later analyses/comparisons, poor quality data can lead to inaccurate conclusions or identifications.

Upload screenshot: raw FastQC quality plot (R1 or R2)

Choose File Screenshot…FastQCRead



Save the downloaded FastQC HTML report in `rawfastqc/` using your naming convention.

2) fastp trimming/filtering (Galaxy)

In Galaxy 1. Run **fastp** on your raw reads 2. Keep defaults unless your instructor specifies otherwise 3. Make sure the output includes the report

Download requirement: Download the **fastp report HTML** and save it in: `fastp/`

2A. fastp version (from Galaxy Job Info):

Galaxy version 1.1.0+galaxy0

2B. What did fastp change? (2–3 sentences)

One thing that changed was the filtering of reads with low quality (14.75%). The percentage of reads that were kept and passed the filters was 85.25%. In general, fastp resulted in the filtering out of low-quality reads.

2C. Name one metric that changed from BEFORE → AFTER in the fastp report (e.g., Q30%, GC%, read length distribution) and state how it changed:

One metric that changed in the fastp report was Q30 which had decreased from the initial 220.015 M to the final 198.704 M.

Upload your fastp HTML report (downloaded from Galaxy)

Choose File SR02FastpReport.html

Selected file: SR02FastpReport.html

Save your fastp HTML in `fastp/` using your naming convention.

3) FASTQC on trimmed reads (Galaxy)

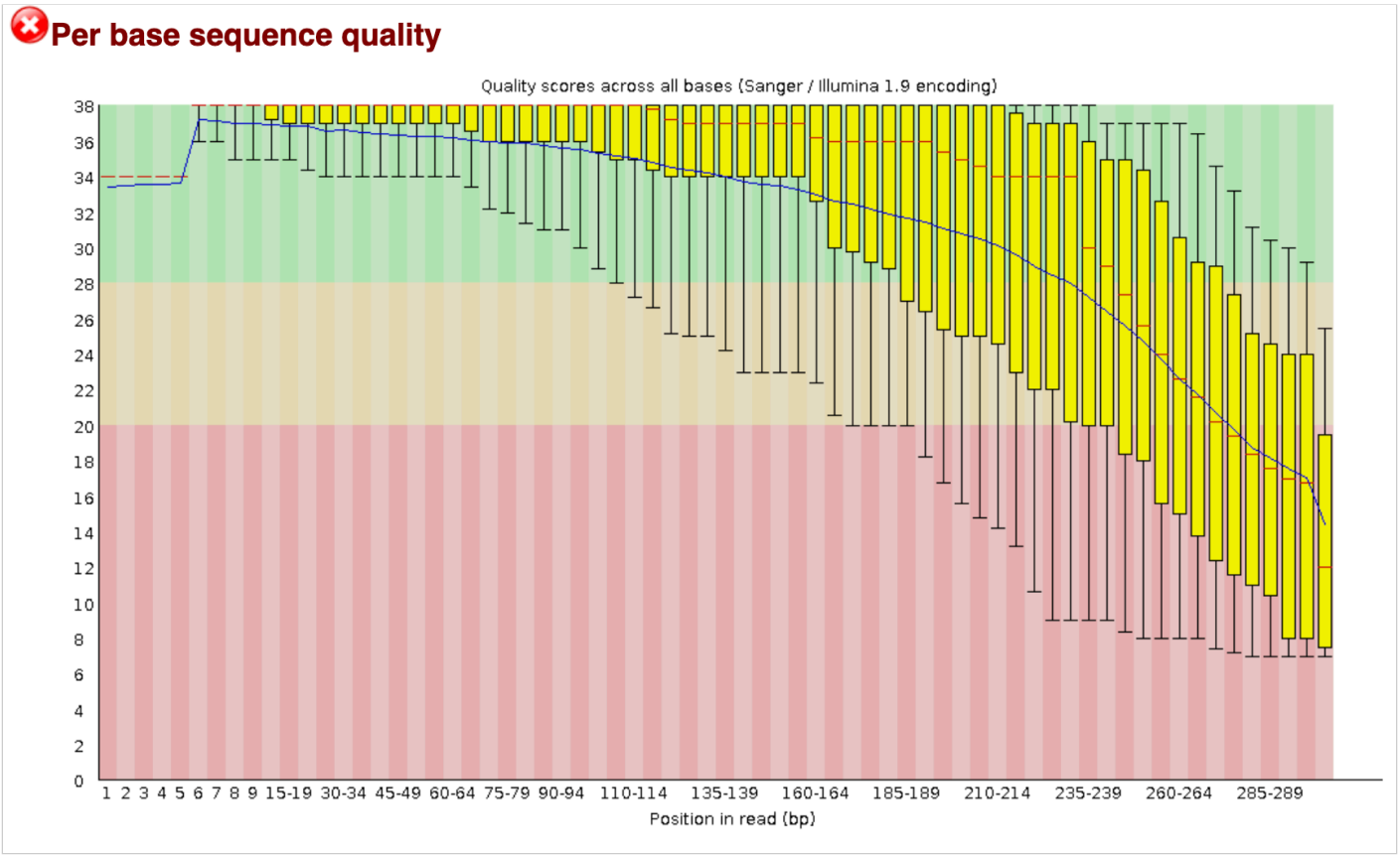
In Galaxy 1. Run **FastQC** on the **trimmed** reads 2. Download the FastQC HTML report for either R1 or R2 and save it in: `trimmedfastqc/`

3A. One key improvement (or remaining issue) after trimming (2–3 sentences):

One improvement after trimming was the slight increase in the per base sequence quality towards the end of the forward read. While there was some improvement after trimming, the end of the read still is of reduced quality.

Upload screenshot: trimmed FastQC quality plot (R1 or R2)

Choose File Screenshot…FastQCRead



Save the downloaded FastQC HTML report in `trimmedfastqc/` using your naming convention.

---

4) Assembly (SPAdes in Galaxy)

In Galaxy 1. Run **SPAdes** on your trimmed reads 2. Download the **contigs FASTA** and save it in: `assembly/`

4A. SPAdes version (from Galaxy Job Info):

Galaxy Version 4.2.0+galaxy0

4B. Name of your downloaded contigs file (after you renamed it):

SR02AssemblyContigs.fasta

---

5) Assembly evaluation (QUAST in Galaxy)

In Galaxy 1. Run **QUAST** on your SPAdes contigs 2. Download the **QUAST PDF report only** and save it in: `quast/`

5A. QUAST version (from Galaxy Job Info):

Galaxy Version 5.3.0+galaxy1

5B. N50:

13012

5C. L50:

161

5D. Largest contig:

58238

5E. Total assembly length:

6773753

Upload your QUAST PDF report (downloaded from Galaxy)

Choose File SR02Quast.pdf

Selected file: SR02Quast.pdf

Save the QUAST PDF in `quast/` using your naming convention.

5F. Based on QUAST, is your assembly fragmented? Use at least 2 metrics to justify (2–4 sentences):

Yes, my assembly is fragmented. One metric that supports this is the number of contigs (938) becuase the more contigs there are, the more fragmented the assembly is. Another metric that supports the idea that this assembly is fragmented is the N50 value (13012). The lower this N50 value is (~13kb), the more fragmented the assembly being observed is.

---

6) Genome quality (CheckM2 in Galaxy)

6A) Filter FASTA (Galaxy)

To reduce tiny contigs, create a filtered assembly for work going forward:

In Galaxy 1. Use **Filter FASTA** 2. Input: your SPAdes contigs FASTA 3. Filter by minimum contig length: **500 bp** 4. Save the filtered contigs as a new dataset in `filteredcontigs`

6A. Confirm your filter rule (one line):

Kept contigs that were 500bp or more

6B) **In Galaxy**

1. Run **CheckM2** on your assembly (contigs)
2. Download the CheckM2 report/output and save it in: `checkm2/`

6B. CheckM2 version (from Galaxy Job Info):

Galaxy Version 1.1.0+galaxy0

6C. Completeness (%):

100.0%

6D. Contamination (%):

0.05%

6E. Interpret these numbers: is this “good enough” for downstream annotation? Why? (2–3 sentences)

I believe that these numbers are "good enough" for downstream annotation. The 100% completeness indicates that the assembly likely has all the necessary genes in within it and the 0.05% contamination shows that there is very little contamination (i.e. another organism's genetic information) within the assembly.

Folder requirement: Put the downloaded CheckM2 report/output in `checkm2/` with your naming convention.

---

7) Prokka annotation (Galaxy)

In Galaxy 1. Run **Prokka** 2. Input: your **filtered** contigs (≥500 bp) 3. Change at least one setting from default as instructed (record it below) 4. Run Prokka

7A. Prokka version (from Galaxy Job Info):

Galaxy Version 1.14.6+galaxy1

7B. What did you change from Prokka defaults? (1–2 sentences)

The only thing that was changed from the Prokka defaults is that “Yes” was selected for forced genbank compliance.

Download requirements (Prokka): download and save these into `prokka/` - `.txt` (summary) - `.gbk` (GenBank format annotation) - `.ffn` (nucleotide sequences of features)

7C. Paste your Prokka summary lines (from the .txt) and describe why this does or does not look like a typical bacterial genome:

organism: Genus species strain / contigs: 938 / bases: 6773753 / CDS: 6046 / gene: 6131 / rRNA: 8 / repeat_region: 3 / tRNA: 76 / tmRNA: 1
This does look like a typical bacterial genome. It has a under 7 million bases (would expect much more if eukaryote), the amount of tRNAs falls within the ~30-100 range, and the number of coding sequences is about the same (typical range for bacteria is ~3,500 - 5,500 CDS).

7D. How many hypothetical proteins does your organism have? What percent of CDS are hypothetical?

3,143 / 51.72%

rRNA feature check (from Prokka outputs)

Open your Prokka outputs and locate the rRNA entries (typically in the `.ffn` and/or `.tbl` / `.gff` depending on Galaxy wrapper).

7E. Paste the description/header lines for 23S, 16S, and 5S rRNA (three lines total):

```
>AMOAFMPE_05549 23S ribosomal RNA
>AMOAFMPE_05953 16S ribosomal RNA
>AMOAFMPE_02796 5S ribosomal RNA
```

7F. Paste the full 16S rRNA FASTA sequence (header + sequence):

```
GCATTAAGTGCCCCGCCTGGGGAGTACGGCCGCAAGGCTAAAACTCAAAGGAATTGACGG
GGGCCCGCACAAGCGGCGGAGCATGTGGCTTAATTCGACGCAACGCGAAGAACCTTACCA
AGGCTTGACATACACCGGAAAGCATCAGAGATGGTGCCCCCCTTGTGGTCGGTGTACAGG
TGGTGCATGGCTGTCGTCAGCTCGTGTCGTGAGATGTTGGGTTAAGTCCCGCAACGAGCG
CAACCCTTGTCCCGTGTTGCCAGCAGGCCCTTGTGGTGCTGGGGACTCACGGGAGACCGC
CGGGGTCAACTCGGAGGAAGGTGGGGACGACGTCAAGTCATCATGCCCCTTATGTCTTGG
GCTGCACACGTGCTACAATGGCCGGTACAATGAGCTGCGATACCGCGAGGTGGAGCGAAT
CTCAAAAAGCCGGTCTCAGTTCGGATTGGGGTCTGCAACTCGACCCCATGAAGTCGGAGT
CGCTAGTAATCGCAGATCAGCATTGCTGCGGTGAATACGTTCCCGGGCCTTGTACACACC
GCCCGTCACGTCACGAAAGTCGGTAACACCCGAAGCCGGTGGCCCAACC
```

---

8) 16S species ID using BLAST (NCBI)

Use your 16S rRNA FASTA from above.

8A. Top hit species name + percent identity + query coverage:

Streptomyces tendae strain ATCC 19812 16S ribosomal RNA, complete sequence / 99.30% identity / 100% coverage

8B. Based on 16S alone, how confident are you in species-level ID? Why? (2–3 sentences)

I am fairly confident that my species-level ID is somewhat accurate. I think that it is a good indicator of the genus (Streptomyces), however, since the 16S rRNA is fairly conserved, I would expect it to be similar to many other organisms.

9) ANI (EZBioCloud) — two comparisons

Use the ANI tool and compare your assembly to **two reference genomes** that you choose (based on your best guess from 16S/Prokka/TYGS). Open the site here (new tab): EZBioCloud ANI (https://www.ezbiocloud.net/tools/ani)

Download requirement: download the two reference genomes you used and save them in `ani/` with clear names.

9A. Reference genome #1 (species/strain):

Streptomyces tendae strain ATCC 19812

9B. ANI % vs reference #1:

99.30%

9C. Reference genome #2 (species/strain):

Streptomyces rubrogriseus strain NBRC 15455

9D. ANI % vs reference #2:

83.11%

9E. Interpret your ANI: do you meet a typical same-species ANI threshold? What conclusion do you draw? (2–4 sentences)

Yes, the ANI did meet the typical same-species ANI threshold of at least 95% for reference genome #1. As a result, it is possible to conclude that they are the same species. This is not the case for reference genome #2 as the ANI percentage was 83.11% (less than that threshold).

10) TYGS phylogeny (use filtered contigs for this report)

Run TYGS using your filtered contigs (≥500 bp). Export/download the tree (.phy file) and view in iTOL. Make sure that you have displayed the bootstrap values. Export your tree as a .pdf file. The .phy and .pdf should be in your `tygs/` folder

In your own words: what kinds of evidence does TYGS use to place your genome? Did you need to generate a proteome tree or was the genome tree sufficient? Did it correspond to what you found with your 16S BLAST results.

TYGS was unable to complete my request by the time of submission. Attached is the confirmation page. TYGS uses the entire genome in addition to features such as the 16S rRNA and compares it to other genomes. If it is of high quality, a genome tree can be sufficient but if there is poor support, then a proteome tree can be best.

Upload your TYGS/iTOL tree as a PDF

Choose File TYGSSubmissionPage

11) Genome mining (Day 14 web tools) — ALL REQUIRED

You must run **all** tools below and include a **screenshot of results** for each. Save screenshots in: `miningfiles/` with clear names.

Also: record the **version/date/database** if the tool reports it (many do on the results page or in the footer).

11A) CARD / RGI (AMR genes)

Open: CARD / RGI (https://card.mcmaster.ca/analyze/rgi)

11A1. CARD/RGI version + database/date info (copy from results page if shown):

RGI Version 6.0.5, CARD Version 4.0.1

11A2. Key AMR findings (2–4 sentences). Include at least one gene + evidence type (e.g., perfect/strict/loose):

One key finding is the gene that encodes HelR which is highly similar to the sequence being referenced (strict). This is a helicase-like RNA polymerase protection protein that binds to rifamycin, allowing transcription to occur (RNA polymerase is not inhibited). Another key finding is is gene that encodes the Streptomyces venezuelae rox (also strict). This allows for resistance to rifamycin by opening the ring of rifamycin, preventing the antibiotic from binding to RNA polymerase.

Upload screenshot: CARD/RGI results

Choose File STendaeCardRgi

Results (all Loose hits shown)

Search:

RGI Criteria	ARO Term	SNP	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity of Matching Region	% Length of Reference Sequence	AST Source
Strict	Streptomyces venezuelae rox		protein homolog model	rifampin monooxygenase	rifamycin antibiotic	antibiotic inactivation	67.01	102.52	
Strict	vanW gene in vanI cluster		protein homolog model	vanW, glycopeptide resistance gene cluster	glycopeptide antibiotic	antibiotic target alteration	30.48	153.08	
Strict	HelR		protein homolog model	helicase-like RNA polymerase protection protein	rifamycin antibiotic	antibiotic target protection	69.54	104.99	

Previous 1 Next

Save as something like `SampleID_card_rgi.png` in `miningfiles/`.

11B) ResFinder (AMR genes)

Open: ResFinder (https://genepi.food.dtu.dk/resfinder)

11B1. ResFinder version/database info (copy from results page if shown):

Version 4.7.2

11B2. Key findings (2–4 sentences). Do ResFinder and CARD agree? If not, why might they differ?

CARD and ResFinder slightly disagree. While they both say that there is resistance, CARD gives three results that confer resistance to rifamycin and glycopeptides whereas ResFinder gives a result related to oleandomycin. They may differ as ResFinder looks for only horizontally acquired resistance genes and CARD looks for those in addition to mutations that may have allowed for resistance.

Upload screenshot: ResFinder results

Choose File STendaeResFinder

Acquired AMR gene hits Hide

Resistance gene	Identity	Contig	Phenotype(s)	PMID	Accession no.
ole(C)	87.88 %	NODE_3_length_48373_cov_23.907847	Oleandomycin	8326867, 27006457	L06249

Save as `SampleID_resfinder.png` in `miningfiles/`.

11C) PathogenFinder 2.0 (pathogenicity prediction)

Open: PathogenFinder (https://genepi.food.dtu.dk/pathogenfinder)

11C1. PathogenFinder version/database info (copy from results page if shown):

Version 0.6.0

11C2. Report the predicted probability/label and interpret it (2–4 sentences):

The predicted probability that the organism is a pathogen to humans is 45.99%. This means that it is likely not going to cause disease, however, it is not absolute and there could be a slight possibility.

Upload screenshot: PathogenFinder results

Choose File STendaePathogenFinder

Module	Prediction
Neural Network 1	0.3945
Neural Network 2	0.522
Neural Network 3	0.3845
Neural Network 4	0.5386
Mean	**0.4599 (0.0707)**

The bacterial genome is predicted to have *Human Non Pathogenic* capacity

Save as `SampleID_pathogenfinder.png` in `miningfiles/`.

11D) VFDB virulence factors via **ABRicate** (Galaxy)

The VirulenceFinder/VFDB web outputs can be inconsistent, so for this report you will run VFDB searches **inside Galaxy** using **ABRicate**.

In Galaxy (ABRicate): 1. Search tools for **ABRicate** 2. **Input:** your **Prokka GenBank file** (`.gbk`) 3. Under**Advanced**, **Database:** choose **VFDB** 4. Run ABRicate 5. Download the **tabular** results file (`.tabular`) and save it in `miningfiles/` using your naming convention.

11D1. ABRicate version + VFDB database version (from Galaxy Job Info):

ABRicate Version: Galaxy Version 1.0.1 / VFDB (2024-Dec-15)

11D2. How many VFDB hits did you get (from the results table/summary)?

0

11D3. List your top 3 VFDB hits (gene name + brief description/function). If you have fewer than three, list all. If you have none, you can put NA:

NA

Upload your ABRicate VFDB results table (tabular)

Choose File STendaeAB…VFDB.tabular

Selected file: STendaeABRicateVFDB.tabular

Save as `SampleID_abricate_vfdb.tabular` in `miningfiles/`.

11E) antiSMASH (biosynthetic gene clusters)

Open: antiSMASH (https://antismash.secondarymetabolites.org/)

11E1. antiSMASH version/database info (copy from results page if shown):

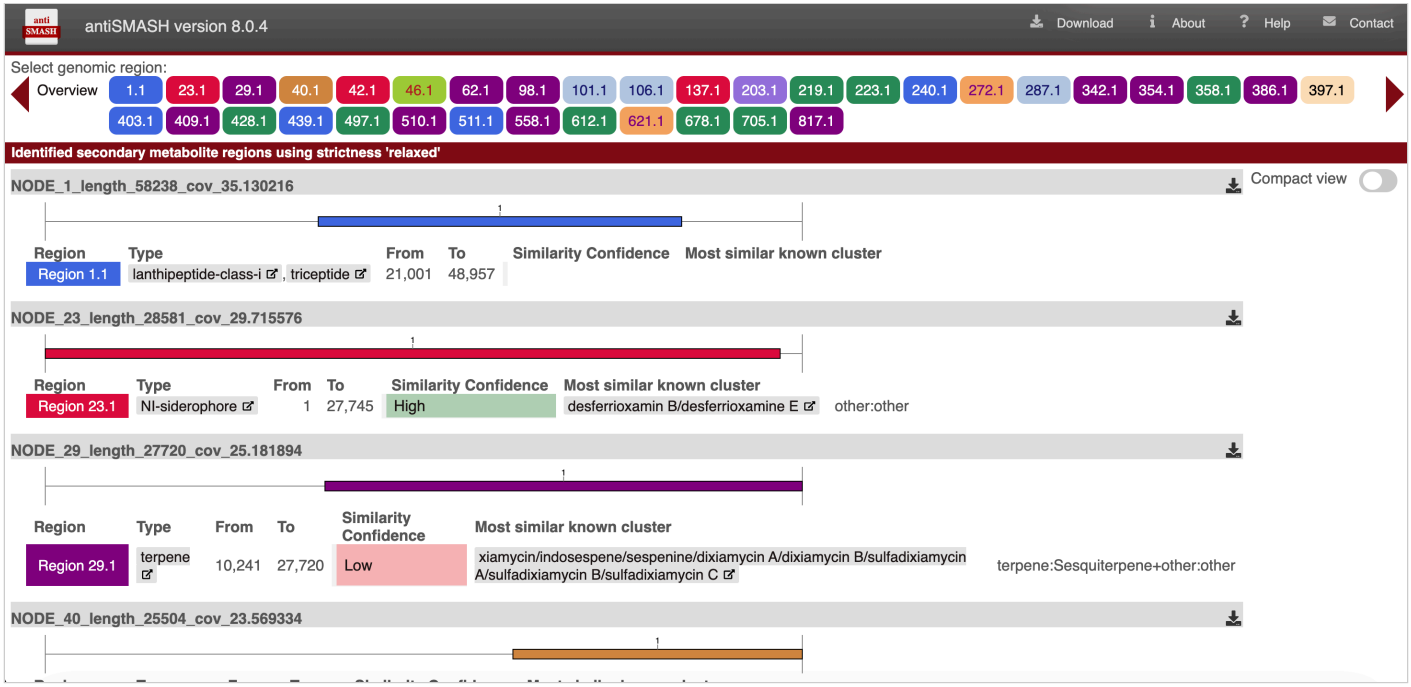
Version 8

11E2. Key BGC finding(s) (2–4 sentences). Include at least one cluster type and whether it matches a known cluster:

There were 35 clusters/regions found in the filtered contigs file. One cluster found is the NI-siderophore (region 23.1) that is going to chelate to nickel ions and bring it back to the microoganism for it to use. It does match a similar cluster called desferrioxamin B/desferrioxamine E that will chelate iron.

Upload screenshot: antiSMASH results

Choose File STendaeAntiSmash



Save as `SampleID_antismash.png` in `miningfiles/` .

11F) IslandViewer (genomic islands)

Open: IslandViewer (https://www.pathogenomics.sfu.ca/islandviewer/)

11F1. IslandViewer version/database info (copy from results page if shown):

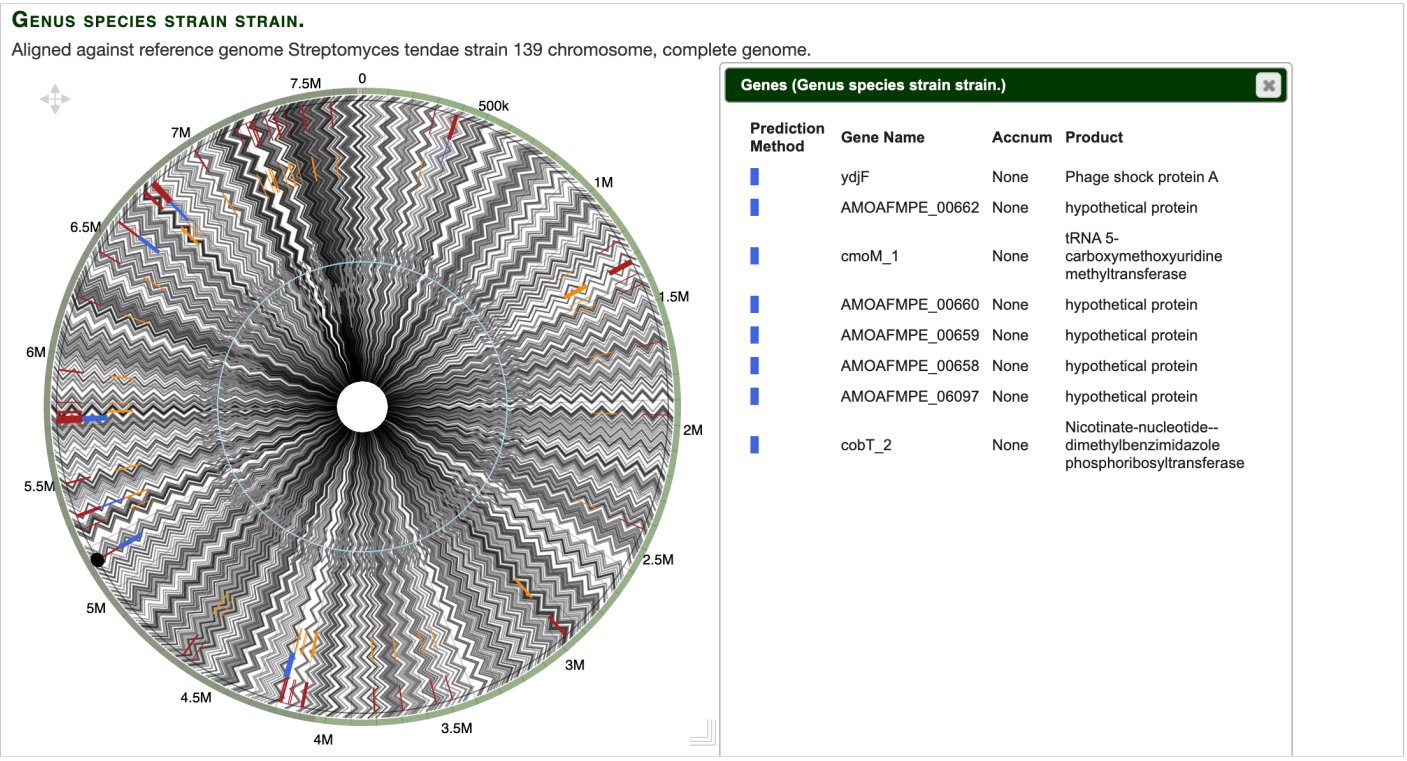
Version 4

11F2. Key island finding(s) (2–4 sentences). How many predicted islands? Any notable features (mobility genes, tRNA insertion sites, etc.)?

There were 322 total predicted islands. One notable feaure is the presence of a tRNA 5-carboxymethoxyuridine methyltransferase which adds a methyl group to a uridine in the tRNA. Another notable feature is the presence of phage shock protein A which is a protein that will help the membrane of the cell be maintained when it undergoes stress.

Upload screenshot: IslandViewer results

Choose File STendaeIslandViewer4



Save as `SampleID_islandviewer.png` in `miningfiles/` .

11G) CRISPRCasFinder (CRISPR arrays + Cas genes)

Open: CRISPRCasFinder (https://crisprcas.i2bc.paris-saclay.fr/)

11G1. CRISPRCasFinder version/database info (copy from results page if shown):

CRISPR-Cas++ 1.1.2

11G2. Key CRISPR/Cas findings (2–4 sentences). Did you find CRISPR arrays? Cas genes? What confidence level?

CRISPRCasFinder said the sequence was >100 and CRSIPRCasMeta did not give a result.

Upload screenshot: CRISPRCasFinder results

Choose File STendaeCRISPRCasMeta

CRISPRCasFinder [online]

Viewing Result

No result

- Submission date : 03/24/2026 19:04:05
- Id : 639099758456864591
- File name : SR02FilteredContigs.fasta
- Job name : No name provided

```
Your search did not give a result.
```

Save as `SampleID_crisprcasfinder.png` in `miningfiles/`.

11H) Synthesis across genome mining tools

11H1. In 6–10 sentences, synthesize what genome mining suggests about this organism (resistance potential, virulence potential, mobile elements, secondary metabolism, etc.). Point to at least one result from each tool:

All things considerd, the genome mining toolds suggest that, while this organism is likely not a human pathogen, it has the potential to cause disease. According to PathogenFinder, this organism is predicted to not have pathogenic capacity in humans (liklihood is 45.99%). VirulenceFinder reiterates this idea as it did not identify any virulence genes in the organism itself. Despite the organism likely not being a pathogen, it still has some qualitis that, when expressed, could help increase pathogenicity. When it comes to antibiotic resistance, CARD/RGI found resistance to rifamycin and glycopeptide whereas ResFinder found resistance to only oleandomycin. These examples of resistance would make the organism more difficult to be removed from the host, however, while these genes are present in the genome, it does not necessarily mean they are expressed. AntiSMASH found 35 clusters responsible for secondary metabolites, including one that results in a Ni-siderophore. This would assist the organism in obtaining nickel when secreted that can then be used in a variety of cellular processes. Lastly, IslandViewer was able to identify a total of 322 islands such as one that contains a gene for a phage shock protein. The products of this would help the organism survive sterssful conditions by maintaining the integrity of its membrane.

CRISPRCasFinder did not provide any results.

12) Final synthesis (your conclusion)

12A. What is your best-supported species ID? Use at least 3 evidence types (16S, ANI, TYGS, CheckM2, Prokka) and explain remaining uncertainty (8–12 sentences):

Based on the data collected, the species ID that is best-supported by 16S, ANI, and CheckM2 is Streptomyces tendae. According to the CheckM2 report, there was 100% completeness and 0.05% contamination. The completeness refers to the idea that all of the necessary genes for microorganisms were found in the genome. Additionally, the low contamination indicates that there is very little contamination from other organisms present and only information pertaining to the organism of interest was present. After using BLAST, the top result from the 16S data was that of a strain of S. tendae (ATCC 19812). This result had a 99.3 percent identity, meaning that 99.3% of 16S sequence used as a query sequence were an exact match with the reference sequence. Another result from BLAST was that there was 100% coverage or alignment of the inputted sequence with the reference sequence. A third piece of evidence that supports this is the results from ANI. When the assembly was compared with the reference genome (S. tendae ATCC 19812), the ANI percentage was 99.30%. This high percentage did surpass the 95% threshold, indicating that the assembly and the reference genome are the same species. After collecting and interpreting all of this data, it is determined that the best-supported species ID is Streptomyces tendae.

Export as PDF

When you are ready: 1. Ensure your screenshots/PDFs are uploaded into the page. 2. Click **Download as PDF (Print)** and save the PDF. 3. Verify that your notes file and your PDF are in the folder to be uploaded. 4. Commit your **folder structure + downloaded reports + renamed files + notes** to Git.